



MARKOV CHAIN MONTE CARLO METHODS

RANDOM NUMBERS AS A TOOL FOR PHYSICS SIMULATIONS

David Lawton

A TPSA Workshop

September 4, 2026



Table of Contents

- ▶ Monte Carlo?
- ▶ Markov Chain?
- ▶ MCMC Algorithm & Statistical Physics Simulations
- ▶ Potts Model
- ▶ A Breakdown? Frustrated Systems
- ▶ Bibliography



Monte Carlo?



Random Numbers for Integration

1 Monte Carlo?

'*Monte Carlo*' methods generally encapsulate the use of random sampling to get numerical results. The simplest example is in numerical integration, a case you should be familiar with. Considering independent random variables $\{X_i\}$ uniformly distributed in $(0, 1)$, we can see that

$$\langle g \rangle = \int_0^1 g(x) dx = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N g(X_i),$$

by the strong law of large numbers. A generalisation exists for independent random variables $\{X_i\}_f$ with some non-uniform distribution $f(x)$. The power of these methods really comes from the equivalent relations for multidimensional integrals and independent variable $\{X_i\}, \{Y_i\}$.



A Classic Example

1 Monte Carlo?

The simplest example is the approximation of π . A point of relevance, in this example, to physics applications is the introduction of the accept/reject logic. In MC physics simulations, the space we wish to integrate over is that of system states, be that over $(\vec{x}, \vec{p})$ -space, field configurations or spin configurations. 2013

$$\begin{aligned}\frac{\pi}{4} &= \int_0^1 dx \sqrt{1-x^2}, \\ &= \int_0^1 dx \int_0^{\sqrt{1-x^2}} dy, \\ &= \int_0^1 dx \int_0^1 dy \underbrace{\theta(1-x^2-y^2)}_{g(x)}, \\ &\simeq \frac{1}{N} \sum_{k=1}^N g(X_k, Y_k) = \frac{N_{\text{in}}}{N_{\text{tot}}}.\end{aligned}$$



Markov Chain?



Defining a Markov Chain

2 Markov Chain?

In physics terms, a *Markov Chain* is a series of random variables $\{X_i\}$ where the probability of X_{k+1} being state i is dependent only on the current state j of X_k . We label these fixed transition probabilities P_{ij} , obeying $\sum_i P_{ij} = 1$. The requirements on P_{ij} are history and time independence

$$\begin{aligned} P\{X_{k+1} = i \mid X_k = j, \{X_{k-l} = j_l\}\} &= P\{X_{k+1} = i \mid X_k = j\} \quad \forall i, j, j_l. \\ P\{X_{k+1} = i \mid X_k = j\} &= P\{X_{n+1} = i \mid X_n = j\} \quad \forall n, k. \end{aligned}$$

We can also define the long-run proportion of time spent in each state π_j which uniquely satisfy

$$\sum_j \pi_j = 1, \quad \sum_j P_{ij} \pi_j = \pi_i.$$



Irreducibility, Aperiodicity and Stationary Probabilities

2 Markov Chain?

A couple of types of Markov chains are of particular interest to us,

1. Irreducible: Given a particular initial state, any other state can be reached with non-zero probability.
2. Aperiodic: Informally, series doesn't get stuck in some cycle of states. Also defined by satisfying

$$P\{X_n = j \mid X_0 = j\} > 0 \text{ and } P\{X_{n+1} = j \mid X_0 = j\} > 0,$$

for some $n \geq 0$ and some j .

If both are satisfied, $\{\pi_j\}$ are interpreted as $\pi_j = \lim_{k \rightarrow \infty} P\{X_k = j\}$.



To Equilibrium and Not Beyond!?

2 Markov Chain?

To consider the n^{th} step, one can think of the probabilities as some matrix acting on the previous step,

$$(\mathbf{Q})_{ij} := P_{ij}; \quad p_i^{(n)} = P\{X_n = i\} = (\mathbf{Q}\mathbf{p}^{(n-1)})_i.$$

Evidently, given initial conditions $\mathbf{p}^{(0)}$, long-run behaviour is governed by

$$\lim_{n \rightarrow \infty} \mathbf{p}^{(n)} = \lim_{n \rightarrow \infty} \mathbf{Q}^n \mathbf{p}^{(0)} = \boldsymbol{\pi}?, \quad \lim_{n \rightarrow \infty} \sum_{i=0}^n h(X_i) = \sum_j \pi_j h(j).$$

If the chain is irreducible the result is independent of i.c. and if aperiodic, the convergence is well-defined, thus the two conditions are required for the final result to be viewed as the equilibrium distribution. Any function $h(j)$ can then be measured at equilibrium!



MCMC Algorithm & Statistical Physics Simulations



Metropolis-Hastings

3 MCMC Algorithm & Statistical Physics Simulations

To introduce our algorithm, we consider proposal irreducible transitions $q(i, j)$ and set, with α acceptance probabilities defined by q ,

$$P_{ij} \underset{i \neq j}{=} \underset{\text{Proposal}}{q(i, j)} \cdot \underset{\text{Acceptance}}{\alpha(i, j)} \ ,$$
$$\underset{i=j}{=} q(i, i) + \sum_{k \neq i} q(i, k)(1 - \alpha(i, k)).$$

It can be shown the chain will be time-reversible have well-defined π if

$$\alpha(i, j) = \min \left(\frac{\pi_j q(j, i)}{\pi_i q(i, j)}, 1 \right) .$$

Metropolis-Hastings is specified by symmetric q .



Algorithm in Practise

3 MCMC Algorithm & Statistical Physics Simulations

So using a Metropolis algorithm, how can observables be measured and how can we measure the error of our samples?

1. Equilibration: Considering an irreducible, aperiodic chain, we require a large N to reach equilibrium. In practise, the required N depends on the system, the **initial conditions** and even the temperature.
2. Use statistical mechanics relations to estimate properties like specific heat/magnetic susceptibility
3. Estimate mean-squared error, either by splitting sample into uncorrelated batches, or by extracting relaxation time via autocorrelations.



Initialisation

3 MCMC Algorithm & Statistical Physics Simulations

An important part of any physics simulation is a reasonable choice of initial conditions. Consider the Ising model, localised spins in d dimensions,

$$H = - \sum_{\langle i,j \rangle} J_{ij} \sigma_i \sigma_j, \quad \sigma_k = \pm 1.$$

In the antiferromagnetic case ($J > 0$), randomly allocated spins is a reasonable guess, but for the ferromagnetic case, it lies much farther from the equilibrium, and much more time is needed to allow for convergence to the correct distribution. In this simple model the consequences are minimal, but in a more complex energy landscape (e.g. glass-forming systems), a poor choice can seriously hurt simulations' quality.



Properties

3 MCMC Algorithm & Statistical Physics Simulations

When simulating a physical system, the aim is usually to compare to experimental data, and for this purpose we require the extrapolation of macroscopic thermodynamic quantities. For indices representing discrete states,

$$E = H(i), \quad \langle E \rangle = \frac{1}{N} \sum_{k=0}^N H(X_k)$$

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = \beta^2 \sigma_E^2$$

$$\chi = \frac{\partial \langle M \rangle}{\partial H} = \beta \sigma_M^2$$

The definition of M is be model dependent. We define σ_A as the sample standard deviation of observable A ,

$$\langle A \rangle := \frac{1}{N} \sum_{k=0}^N A(X_k)$$

$$\sigma_A^2 := \langle A^2 \rangle - \langle A \rangle^2$$

In MC simulations, σ_A cannot be used as standard error of the observables, as successive steps in the chain are correlated.



Error

3 MCMC Algorithm & Statistical Physics Simulations

Instead of standard deviation we must calculate standard error from *mean-squared error*, which we can define for correlated or uncorrelated samples,

$$\text{Uncorrelated: } \text{MSE}(A) = \frac{1}{N_s} \sum_s \sigma_{A,s}^2,$$

$$\text{Correlated: } \text{MSE}(A) = \frac{1}{N_{\text{eff}}} \sum_s \sigma_{A,s}^2, \quad N_{\text{eff}} = N/\tau_A$$

Either can be used, the first by splitting your chain into approx. uncorrelated batches, the second by estimating the time for A to decorrelate, via integrated relaxation time τ_A . This is done by integrating the autocorrelation of A

$$\tau_A = \sum_{l=0}^{\infty} C_A(l)/C_A(0),$$



Potts Model



Standard Potts Model

4 Potts Model

The *Potts model* is a generalization of the Ising model, where each site has q states ($q = 2$ is Ising), with a Hamiltonian

$$H(s) = - \sum_{\langle i,j \rangle} J_{ij} \delta(s_i, s_j), \quad s_i = 1, \dots, q, \quad \delta(s_i, s_j) = \begin{cases} 1, & s_i = s_j \\ 0, & s_i \neq s_j \end{cases}.$$

The sites lie on a d -dim. $L_1 \times \dots \times L_d$ periodic square lattice, where we can consider i a d -dim vector of indices. s specifies the state of the system. The equilibrium distribution is given by the Boltzmann distribution

$$\pi(s) = \frac{e^{-\beta H(s)}}{Z}, \quad Z = \sum_s e^{-\beta H(s)}, \quad \beta = \frac{1}{T}$$



Useful Properties

4 Potts Model

There are some properties it will be handy to define,

1. Fractional Occupancy: We can denote the proportion of spins in each state $\lambda = 1, \dots, q$ by $x_\lambda(s)$. We can then use these to create an order parameter,
2. Fractional Magnetization: By taking the maximum occupancy fraction, we can define fractional magnetisation $M(s)$ by

$$M(s) = \frac{q \cdot \max(x_\lambda(s)) - 1}{q - 1},$$

which can be used as an order parameter, and used to find χ . At high T , $x_\lambda \rightarrow 1/q \Rightarrow M(s) \rightarrow 0$, but when the system cools and a particular λ dominates more, $M(s) \rightarrow 1$.



Metropolis Algorithm for Potts

4 Potts Model

So how can we implement the Metropolis algorithm? Given an initial state $X_0 = s$, we shall consider some change $s \rightarrow s'$, given by $s_k \rightarrow s'_k \neq s_k$, with k sequentially sweeping through the lattice. In this way $q(s', s)$ can be defined as

$$q(s', s) = \begin{cases} \frac{1}{q-1}, & s'_k \neq s_k \\ 0, & s'_k = s_k \end{cases},$$

which is symmetric, $q(s', s) = q(s, s')$. Our acceptance probability α is thus defined by

$$\alpha(s', s) = \min \left(\frac{\pi(s')}{\pi(s)}, 1 \right) = \min \left(e^{-\beta \Delta H(s', s)}, 1 \right)$$



Implementation

4 Potts Model

We can now write a ‘recipe’ to follow for our simulations,

1. Consider lattice site i .
2. Propose random choice of $s'_i \neq s_i$.
3. Calculate change in energy $\delta = \Delta H(s', s)$ and if
 - 3.1 $\delta \geq 0$: Let $s_i = s'_i$
 - 3.2 $\delta \leq 0$: Generate random number $p \in (0, 1)$. If $p \leq e^{-\beta\delta}$, let $s_i = s'_i$, else leave state unchanged.

One then repeats the process for $L_1 \times \cdots \times L_d$ sites, via either a random or sequential ‘sweep’. Perform N sweeps, collecting observable data after each.



Exercises

4 Potts Model

The first exercise is to

- Construct a program to execute the Metropolis algorithm given the system parameters $\{d, q, \beta, J, \{L_i\}, N\}$. Assume a $d = 2$ if needs be.

Choose your own initialisation method, and decide on either random or sequential sweeps, but try to comment & compartmentalise your code to make it easier for you (or someone else) to read or edit your code. Have your program collect and output your (E, M) data for all sweeps preferably to some .csv file for post-analysis.



- Run a $d = 2$ simulation for some $J > 0$, some $q \leq 4$, and some value of L on the order of 10. Choose a value of β above the critical point $\beta_c = \frac{1}{J}(1 + \sqrt{q})$.
 - Examine the behaviour over time of the outputted observables to inform future runs, and run for varying choices of initialisation and sweeping for comparison.
- Write a program to process the simulation data and output sample estimations of equilibrium E, M, C_V , and χ .
- Extend this program to calculate E and M autocorrelations from the simulation data, and output their relaxation times τ_A . Use this to estimate the mean-squared error as discussed.



Autocorrelations & Relaxation Time

4 Potts Model

For some system variable A , the autocorrelation function is defined as

$$C_A(l) = \sum_k \langle A_k A_{k+l} \rangle - \langle A \rangle^2 \sim e^{-l/\tau_A}$$

where l represents time in MC steps, and τ is the relaxation time. We can estimate τ_A by integrating the normalised autocorrelation over l .

$$\tau_A = \sum_{l=-\infty}^{\infty} \frac{C_A(l)}{C_A(0)} \simeq 1 + 2 \sum_{l=1}^M \frac{\tilde{C}_A(l)}{\tilde{C}_A(0)} =: \tilde{\tau}_A(M) \xrightarrow{M \rightarrow \infty} \infty$$

where $\tilde{C}_A(l)$ is the sample autocorrelation, which will cause a noisy τ_A estimation if too large. One must in practise calculate τ_A for many M to find the minimum M satisfying $M \leq 5\tilde{\tau}_A(M)$.



- Run the simulation over a range of β around the critical point. Plot $E, M, C_V, \chi, \tau_E, \tau_M$ against β . Try to determine critical behaviours of these properties and what they may say about the efficacy of our methods in finding accurate results, and the physical behaviour of the system.
- Run the same for $q = 5$ and determine the changes in behaviour.



A Breakdown? Frustrated Systems



Frustrated Physics (Own Medecine?)

5 A Breakdown? Frustrated Systems

Frustrated systems are systems in which varying, competing interactions make it difficult for the system to find its ground state. There are two main sources of ‘frustration’, system geometry, and varying spin couplings. It is a relatively simple process to introduce frustration to the Potts model, and you can choose to do it in either way

- Geometric Frustration (Ice): Drop down to $q = 2$, set $J < 0$, and build a kagome or triangular lattice for $d = 2$, or pyrochlore or diamond lattice for $d = 3$. Note the existence of fractional excitations!
- Frustrating Couplings (Glass): Where our original model sums over neighbours with a fixed global coupling, the frustrated model will randomly sample J_{ij} from $\{-1, +1\}$ or a Gaussian distribution for all i, j while building the system.



Exercise

5 A Breakdown? Frustrated Systems

- Choose a type of frustration and build initialisation, energy and sweeping methods.
- Simulate the system in the same way as the previous one. Vary your system size and steps as you see fit to get reasonable results without increasing computation time too much. Compare your results to non-frustrated models.
- Further frustration specific exercises:
 - Geometric: Vary q or try the higher dimensional model. Add opposing next-nearest neighbour coupling.
 - Coupling-induced: Vary q or draw couplings from another distribution (Levy, ...).



Thank you for your attention!
Any questions?



Bibliography



Bibliography

7 Bibliography

-  (2013). In: *Simulation (Fifth Edition)*. Fifth Edition. Academic Press, pp. 303–310. ISBN: 978-0-12-415825-2. DOI: <https://doi.org/10.1016/B978-0-12-415825-2.00018-8>. URL: <https://www.sciencedirect.com/science/article/pii/B9780124158252000188>.
-  Me (n.d.). In: ().